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ElderGuardian

Smart Home Ecosystem for Elderly Care with AI & Robotic Assistance

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Project Overview

ElderGuardian+ is an AI-powered smart elderly care system designed to improve the safety, health, and quality of life of senior citizens who live independently while keeping their families and healthcare providers continuously connected to their daily well-being.

As people age, they often require regular medication, continuous health monitoring, and immediate assistance during emergencies. However, many elderly individuals prefer living independently rather than in assisted care facilities, which increases the risk of unnoticed medical incidents, medication delays, and reduced daily activity.

ElderGuardian+ addresses these challenges by combining **Artificial Intelligence (AI), Internet of Things (IoT), Robotics, Mobile Technologies, and Web Applications** into one integrated healthcare platform.

Unlike traditional elderly monitoring systems that rely on cameras and continuous visual surveillance, ElderGuardian+ focuses on **environmental awareness rather than personal surveillance**. The system analyzes environmental data, daily routines, activity patterns, medication adherence, and sensor readings to understand the elderly person's condition while preserving their privacy.

The system combines:

- **A Smart Home Assistant Robot** that interacts directly with the elderly person through voice reminders, wellness checks, medication assistance, emergency support, and automatic self-charging using a Dock Station.
- **Smart Environmental Sensors** that monitor activity levels, room movement, sleep patterns, environmental conditions, and home entry/exit events without using cameras.
- **A Mobile Application** that enables family members to monitor their loved one's daily status, receive real-time notifications, communicate with the robot, and manage medications remotely.
- **A Web Platform** that allows doctors and administrators to monitor patients, review health reports, manage medications, schedule appointments, and provide medical recommendations.
- **An AI Analytics Engine** that analyzes behavioral patterns, medication adherence, activity trends, sleep quality, and environmental data to detect abnormal situations and generate intelligent recommendations.

Unlike traditional monitoring systems, ElderGuardian+ does not depend on cameras or constant visual surveillance. Instead, it analyzes environmental patterns and daily routines to provide privacy-preserving care.

1. Robot (Home Assistant)

The **Home Assistant Robot** serves as the primary interaction point between the elderly person and the ElderGuardian+ system. It is designed to be simple, user-friendly, and affordable, allowing elderly users to interact with the system without the need for smartphones or wearable devices.

Rather than performing complex processing locally, the robot acts as an intelligent communication device connected to the cloud-based backend and AI engine. This design reduces hardware complexity while allowing continuous software improvements and future feature enhancements.

The robot is capable of moving autonomously inside the home, communicating with the elderly user through voice and visual notifications, collecting sensor data, and responding to commands from family members and healthcare providers.

Robot Components

The robot consists of the following hardware components:

- ESP32 Microcontroller
- OLED Display Screen
- Speaker Module
- LED Status Indicator
- Two Encoder DC Motors
- Motor Driver
- Rechargeable Battery Pack
- Automatic Charging Dock (Dock Station)
- Infrared (IR) Dock Detection Module
- Medication Confirmation Button
- Wellness Response Buttons
- Emergency (SOS) Button

Robot Functionalities

1. Morning Assistant

The robot starts the elderly person's day with a friendly greeting based on detected daily activity.

When the motion sensor detects the user's first movement in the morning, the robot provides:

- *Daily greeting*
- *Current time and date*
- *Medication schedule*
- *Upcoming appointments*
- *Weather reminder (optional)*

Example:

"Good morning. Today is Monday. It is 8:00 AM. Your first medication is scheduled at 9:00 AM."

This feature helps establish a consistent daily routine and encourages medication adherence.

2. Medication Reminder

One of the robot's primary responsibilities is assisting with medication adherence.

At the scheduled medication time, the robot:

- *Plays a notification sound*
- *Displays the medication name on the screen*
- *Speak a voice reminder*

Example:

"It is time to take your blood pressure medication."

After several minutes:

"Have you taken your medication?"

*The elderly user can confirm by pressing the **Medication Confirmation Button**. Notify family*

If no confirmation is received:

- *The reminder is repeated.*
- *A medication delay is recorded.*
- *Family members receive a notification.*
- *Repeated delays are reported to the assigned doctor through the AI analytics system.*

3. Wellness Check

The robot continuously cooperates with the environmental sensors and AI engine to detect unusual inactivity or abnormal daily behavior.

If prolonged inactivity is detected, the robot initiates a wellness check.

Example:

"Are you okay?"

The robot provides two simple response buttons:

- Yes
- Need Help

If the user selects **I'm Fine**, the event is recorded and normal monitoring continues.

If the user selects **Need Help**, an emergency alert is immediately sent to family members.

If no response is received within a predefined period, the robot automatically activates the emergency workflow.

4. Family Voice Messages

Family members can communicate with the elderly person remotely through the mobile application.

Instead of requiring live calls, they can send short text messages such as:

Example:

"Did you eat lunch today?"

The backend converts the text into speech using Text-to-Speech (TTS), and the robot plays the message naturally through its speaker.

This feature provides emotional support while keeping communication simple for elderly users.

5. Remote Commands

Authorized family members can remotely interact with the robot through the mobile application.

Available remote actions include:

- Move robot
- Play voice message
- Start a wellness check
- Request the robot to return to the charging dock

6. Emergency Mode

The robot includes a dedicated **SOS Emergency Button** for immediate assistance.

When the SOS button is pressed:

- Emergency notifications are sent instantly to family members.
- The assigned doctor is notified.
- The emergency event is stored in the system.
- The robot displays an emergency status message.
- Audible and visual emergency indicators are activated.

7. Companion Mode

Beyond healthcare monitoring, the robot also provides simple social interaction to reduce feelings of loneliness.:

At appropriate times during the day, the robot may ask friendly questions such as:

- “Would you like water?”
- “Do you need anything?”

This creates emotional support without complexity.

8. Automatic Self-Charging (Dock Station)

To eliminate the need for the elderly person to manually recharge the robot, ElderGuardian+ includes an automatic charging system.

The robot continuously monitors its battery level.

When the battery reaches a predefined threshold:

- The robot suspends non-essential activities.
- It automatically navigates back to its Dock Station.
- The docking position is detected using an Infrared (IR) beacon together with movement tracking.
- The robot aligns itself with the charging contacts.
- Charging starts automatically without user intervention.

Once fully charged, the robot resumes normal operation.

This feature ensures continuous availability of the robot while reducing maintenance requirements for elderly users.

2. Sensors (Environmental Monitoring)

The ElderGuardian+ system relies on a set of smart environmental sensors to understand the elderly person's daily routine without using cameras or intrusive monitoring devices.

Rather than observing the person directly, the sensors continuously collect environmental data such as movement, room occupancy, sleep patterns, home entry and exit events, and surrounding conditions. These readings are transmitted to the ESP32 controller and then forwarded to the AI analytics engine, where they are analyzed to identify behavioral patterns and detect abnormal situations.

Instead of making decisions based on a single sensor, the AI combines data from multiple sensors to improve accuracy and reduce false alarms.

Motion Sensor (PIR)

Purpose : The Passive Infrared (PIR) Motion Sensor detects human movement by sensing changes in infrared radiation.

Functions:

- Detect daily movement
- Measure activity level
- Detect prolonged inactivity
- Track room occupancy
- Support abnormal behavior detection

Examples:

If the AI detects significantly lower movement than the user's normal daily routine, the robot initiates a wellness check.

Example:

Low activity detected → Robot asks: "Are you okay?"

Ultrasonic Distance Sensor

Purpose: The Ultrasonic Sensor measures the distance between the robot and nearby objects.

Functions:

- Detect nearby obstacles
- Assist robot navigation
- Prevent collisions during movement
- Support abnormal event detection

Example:

If the measured distance changes suddenly from:

150 cm



40 cm suddenly

while the motion sensor simultaneously detects unusual movement followed by inactivity, the AI may classify the event as a **Possible Fall**, triggering a wellness check before notifying family members.

The system does not rely on the ultrasonic sensor alone but combines multiple sensor readings to improve reliability.

Door Sensor

Purpose: The Magnetic Door Sensor monitors home entry and exit events.

Functions:

- Detect when the main door opens or closes
- Monitor daily outdoor activity
- Detect unusual absence from home
- Support routine analysis

Example:

Door opened → No Return Detected for an Unusual Period → Family Notification.

This helps family members identify situations where the elderly person may have forgotten to return home or has been away longer than expected.

Bed Sensor

The Bed Pressure Sensor monitors sleeping behavior and rest quality

- Measure sleep duration
- Track bedtime and wake-up time
- Analyze daily sleep patterns
- Support AI wellness analysis

Example

The AI compares daily sleep patterns over time.

If sleep duration decreases significantly or frequent interruptions occur for several consecutive nights, the system recommends a medical follow-up through the doctor dashboard.

Emergency Button

The Emergency Button allows the elderly person to request immediate assistance at any time.

Functions

- Trigger instant emergency alerts
- Notify family members
- Notify the assigned doctor
- Record emergency events in the system
- Activate the robot's emergency mode

Example

One Button Press → Emergency Alert Generated → Family Notification → Doctor Notification

→ Emergency Event Logged

Sensor Integration

Rather than depending on individual sensor readings, ElderGuardian+ combines data from all environmental sensors to create a more accurate understanding of the elderly person's daily behavior.

For example:

- **Low activity** detected by the PIR sensor
- **No response** during the robot's wellness check
- **No door activity**
- **Abnormal sleep pattern**

3. Mobile Application (Family Platform)

The **Family Mobile Application** serves as the primary communication and monitoring platform for family members. Developed using **Flutter**, it enables users to remotely monitor the elderly person's daily activities, receive real-time notifications, communicate with the robot, manage medications, and stay connected with healthcare providers.

The application provides a simple and user-friendly interface that allows family members to quickly understand the elderly person's current condition and respond to important events whenever necessary.

Dashboard

The dashboard provides an overview of the elderly person's current status.

It displays:

- Safety Status
- Last Activity
- Medication Status
- Robot Status
- Battery Level
- Daily Summary
- Latest AI Recommendation

This screen allows family members to monitor the overall situation at a glance.

Live Activity Timeline

The application records important daily events in chronological order.

Example Timeline:

- 08:00 AM — Wake Up Detected
- 09:00 AM — Medication Confirmed
- 11:30 AM — No Movement Detected
- 12:00 PM — Wellness Check Completed
- 02:00 PM — Lunch Reminder Delivered

This timeline helps family members understand the elderly person's daily routine and identify unusual behavior.

Elder Profile

The Elder Profile contains all essential personal and medical information.

It includes:

- Personal Information
- Medical History
- Chronic Diseases
- Emergency Contacts
- Daily Routine
- Assigned Doctor
- Current Medications

This information helps both family members and healthcare providers monitor the patient's condition effectively.

Medication Center

The Medication Center allows authorized users to manage medication schedules.

Features include:

- Add Medication
- Edit Medication
- Delete Medication
- Schedule Medication Reminders
- View Medication History
- Monitor Medication Adherence

Medication reminders are automatically synchronized with the Home Assistant Robot.

Health Monitoring

The Health Monitoring section visualizes the elderly person's daily wellness data.

It displays:

- Daily Activity Level
- Sleep Duration
- Sleep Quality Trends
- Wellness Score
- Activity Statistics
- Environmental Status
- Alert History

This information is generated using data collected from the environmental sensors.

AI Insights

The AI Analytics Engine continuously analyzes behavioral patterns and presents useful recommendations.

Examples include:

- "Daily activity decreased by 25%."
- "Medication adherence has declined this week."
- "Sleep quality has worsened during the past three days."
- "Doctor follow-up is recommended."

These insights help family members identify potential health concerns before they become emergencies.

Family Management

The application supports multiple family members with different permission levels.

Features include:

- Add Family Members
- Manage User Roles
- Shared Notifications
- Emergency Contact Management

This allows several authorized relatives to monitor and assist the elderly person simultaneously.

Remote Robot Control

Family members can remotely interact with the Home Assistant Robot.

Available actions include:

- Move the Robot
- Send Voice Messages
- Start a Wellness Check
- Activate Emergency Mode
- Flash the Emergency LED
- Send the Robot Back to the Dock Station

These features allow family members to assist the elderly person even when they are away from home.

Voice & Communication

The communication module strengthens the connection between the elderly person, family members, and healthcare providers.

Features include:

- Voice Calls
- Recorded Voice Messages
- Text-to-Speech Messages
- Emergency Communication

Family members can send supportive voice messages that the robot plays through its speaker.

Emergency Center

The Emergency Center provides quick access to emergency actions.

Available options include:

- Call Family Members
- Contact the Assigned Doctor
- Call Emergency Services
- View Active Emergency Alerts

This screen is designed for rapid response during critical situations.

Doctor Communication

The application enables secure communication between family members and healthcare providers.

Functions include:

- Chat with Doctor
- Receive Medical Instructions
- Appointment Updates
- Treatment Recommendations

All medical updates are synchronized with the doctor's web platform.

Calendar

The integrated calendar helps organize healthcare activities.

It includes:

- Doctor Appointments
- Medication Schedules
- Follow-up Visits
- Medical Reminders

This ensures that important healthcare events are not missed.

Reports

Family members can generate detailed reports about the elderly person's condition.

Available reports include:

- Daily Reports
- Weekly Reports
- Monthly Reports
- Medication Reports
- Activity Reports
- Sleep Reports

Reports can be reviewed directly within the application and shared with healthcare providers if needed.

Smart Notifications

The application delivers real-time notifications whenever important events occur.

Notification types include:

- Fall Risk Alert
- Medication Delay
- Environmental Warning
- Low Activity Alert
- Emergency Alert
- Doctor Message
- Robot Offline Notification
- Low Robot Battery Notification

Notifications are delivered instantly, allowing family members to respond quickly whenever assistance is needed.

Data Synchronization

All application data is synchronized with the cloud-based backend in real time.

Any updates made by family members, doctors, or the Home Assistant Robot are automatically reflected across the entire ElderGuardian+ platform, ensuring that every authorized user always has access to the latest information.

4. Website (Doctor Platform)

The **Web Platform** serves as the professional management and monitoring system for doctors and administrators. Developed using **Angular**, it provides healthcare professionals with comprehensive access to patient information, AI-generated insights, medical records, appointments, reports, and emergency alerts.

Unlike the mobile application, which is designed for family members, the web platform focuses on medical decision support, patient management, and healthcare administration.

Doctor Dashboard

The Doctor Dashboard provides a complete overview of the doctor's patients and daily activities.

It displays:

- Today's Appointments
- Active Patients
- Critical Alerts
- Recent Notifications
- Patients Requiring Follow-up
- Medication Adherence Overview
- Daily Activity Statistics
- AI Recommendations Summary

This dashboard enables doctors to quickly identify patients who require immediate attention.

Patient Monitoring

Doctors can access a complete profile for each patient.

The monitoring page includes:

- Personal Information
- Medical History
- Chronic Diseases

- Current Medications
- Assigned Family Members
- Activity History
- Sleep Analysis
- Medication Adherence
- Wellness Score
- Sensor Activity Logs
- Alert History

The system continuously updates patient information using real-time data collected from the robot and environmental sensors

AI Health Analysis

The AI engine continuously analyzes patient behavior and generates intelligent recommendations.

The AI detects:

- Behavioral Changes
- Reduced Daily Activity
- Medication Non-Adherence
- Sleep Pattern Changes
- Environmental Risks
- Increased Health Risks

Example Recommendations:

"Medical review recommended."

"Daily activity has significantly decreased during the last five days."

"Medication adherence has dropped below the normal level."

These recommendations help doctors identify health issues at an early stage.

Appointment Management

Doctors can efficiently manage patient appointments.

Available features include:

- Schedule Appointments
- Modify Existing Appointments
- Cancel Appointments
- Send Appointment Reminders
- View Appointment Calendar

Any appointment updates are automatically synchronized with the family mobile application.

Medication Management

Doctors have complete control over patient medication plans.

Available functions include:

- Add New Medication
- Modify Dosage
- Change Medication Schedule
- Pause Medication
- Resume Medication
- Stop Medication
- Review Medication History

All medication updates are instantly synchronized with the robot and family application.

Emergency Center

The Emergency Center displays all high-priority incidents requiring immediate medical attention.

It includes:

- Emergency Alerts
- Fall Risk Alerts
- Medication Delay Alerts
- Abnormal Activity Alerts
- Recent Incidents
- Emergency Response History
- Follow-up Status

This enables doctors to monitor critical situations efficiently.

Doctor Notes

Doctors can maintain detailed medical notes for every patient.

They can record:

- Medical Instructions
- Treatment Plans
- Health Observations
- Follow-up Recommendations
- Lifestyle Advice

New notes are immediately available to authorized family members through the mobile application.

Reports & Analytics

The reporting module provides detailed healthcare analytics.

Available reports include:

- Daily Reports
- Weekly Reports
- Monthly Reports
- Progress Summaries
- Medication Adherence Reports
- Sleep Reports
- Activity Reports
- Alert Reports

Reports can be exported as PDF files for medical documentation.

Patient History

The platform maintains a complete medical timeline for every patient.

Doctors can review:

- Previous Alerts
- Medication Changes
- Appointment History
- Doctor Notes
- AI Recommendations
- Daily Activity Records
- Sleep History

This allows healthcare providers to monitor long-term patient progress.

Data Visualization

To simplify medical analysis, the system presents patient information using interactive charts and graphs.

Doctors can visualize:

- Daily Activity Trends
- Medication Adherence Rate
- Sleep Duration
- Wellness Score
- Number of Emergency Alerts
- Activity Comparison by Week or Month

Visual analytics help doctors recognize long-term behavioral changes more easily.

Notification Center

The platform includes a centralized notification system.

Doctors receive notifications for:

- New Emergency Alerts
- Medication Delays
- High Fall Risk
- Appointment Updates
- AI Recommendations
- Messages from Family Members

Notifications can be filtered based on priority.

Search & Patient Management

Doctors can quickly locate patients using advanced search and filtering tools.

Search options include:

- Patient Name
- Medical Condition
- Assigned Doctor
- Wellness Score
- Alert Status
- Medication Adherence

This simplifies patient management, especially when monitoring a large number of patients.

Administration Panel

The platform includes an administration module for system management.

Administrators can:

- Manage Doctor Accounts
- Manage Family Accounts
- Manage Elderly User Accounts
- Assign Doctors to Patients
- Manage User Roles and Permissions
- View System Statistics
- Monitor Connected Robots
- Monitor Connected Sensors
- Review System Logs

This module ensures secure and efficient management of the entire ElderGuardian+ platform.

Security & Access Control

The web platform implements secure authentication and role-based authorization.

Features include:

- Secure Login
- Role-Based Access Control
- JWT Authentication
- Audit Logs
- Activity Tracking

Only authorized users can access sensitive medical information.

Cloud Synchronization

All updates made through the web platform are synchronized in real time with:

- Home Assistant Robot
- Family Mobile Application
- AI Analytics Engine
- Database

This ensures that doctors, family members, and the robot always operate using the latest available information.

5. Technologies

ElderGuardian+ combines IoT, Artificial Intelligence, Mobile Development, Web Development, and Cloud-Based Technologies to provide a complete smart healthcare solution for elderly care.

Each technology was carefully selected to ensure reliability, scalability, affordability, and ease of implementation.

Hardware Components

The hardware layer consists of a mobile robot, environmental sensors, and an automatic charging system.

ESP32 Microcontroller

The ESP32 serves as the central controller of the robot.

Responsibilities include:

- Reading sensor data
- Controlling robot movement
- Communicating with the backend through Wi-Fi
- Receiving remote commands
- Sending real-time status updates

Environmental Sensors

The robot works together with multiple sensors to monitor the surrounding environment.

The sensors include:

- PIR Motion Sensor
- Ultrasonic Distance Sensor
- Temperature Sensor
- Magnetic Door Sensor
- Bed Pressure Sensor
- Emergency SOS Button

These sensors provide environmental data used by the AI engine to analyze the elderly person's daily behavior.

Robot Hardware

The Home Assistant Robot includes:

- ESP32 Controller
- OLED Display
- Speaker Module
- LED Status Indicator
- Two Encoder DC Motors
- Motor Driver
- Rechargeable Battery Pack
- Medication Confirmation Button
- Wellness Response Buttons
- Emergency SOS Button

These components allow the robot to move, communicate with users, display information, and respond to emergency situations.

Automatic Charging System

To ensure continuous operation, the robot includes an automatic charging mechanism.

Components include:

- Dock Station
- Infrared (IR) Dock Detection Module
- Charging Contacts
- Battery Charging Circuit
- Battery Monitoring Module

When the battery level becomes low, the robot automatically returns to the Dock Station, aligns itself, and starts charging without requiring user intervention.

Backend Development (ASP.NET Core)

The backend is developed using ASP.NET Core (.NET) and serves as the central communication layer between all system components.

Responsibilities include:

- RESTful API Development
- Authentication & Authorization (JWT)
- Business Logic
- Robot Communication
- Sensor Data Processing
- Notification Management
- Medication Management
- Appointment Management
- AI Integration
- Report Generation
- Real-Time Data Synchronization

The backend ensures secure communication between the robot, mobile application, web platform, and database.

Web Application (Angular)

The doctor and administration platform is developed using Angular.

It provides healthcare professionals with tools for monitoring and managing patients.

Main features include:

- Doctor Dashboard
- Patient Monitoring
- AI Health Insights
- Appointment Management
- Medication Management
- Reports & Analytics
- Emergency Center
- Patient History
- Administration Panel

The responsive interface enables doctors to access patient information from any modern web browser.

Mobile Application (Flutter)

The family application is developed using Flutter, allowing deployment on both Android and iOS devices from a single codebase.

The application includes:

- Family Dashboard
- Live Activity Timeline
- Medication Tracking
- AI Insights
- Robot Remote Control
- Emergency Center
- Voice Communication
- Smart Notifications
- Doctor Communication
- Health Reports
- Calendar Management

Flutter provides a smooth user experience while simplifying cross-platform development.

Database (SQL Server)

The system uses Microsoft SQL Server as the primary relational database.

The database stores:

- User Accounts
- Elderly Profiles
- Doctor Profiles
- Family Members
- Medication Records
- Sensor Logs
- Robot Status
- Activity History
- Sleep Records
- AI Recommendations
- Emergency Events
- Reports
- Appointment Records

Artificial Intelligence & Analytics

The AI engine transforms raw sensor readings into meaningful healthcare insights.

AI Techniques

The system applies several AI techniques, including:

- Behavioral Pattern Analysis
- Rule-Based Decision Engine
- Activity Trend Analysis
- Anomaly Detection
- Wellness Score Calculation
- Medication Adherence Analysis
- Sleep Pattern Analysis
- Environmental Risk Analysis

AI Functions

The AI engine is responsible for:

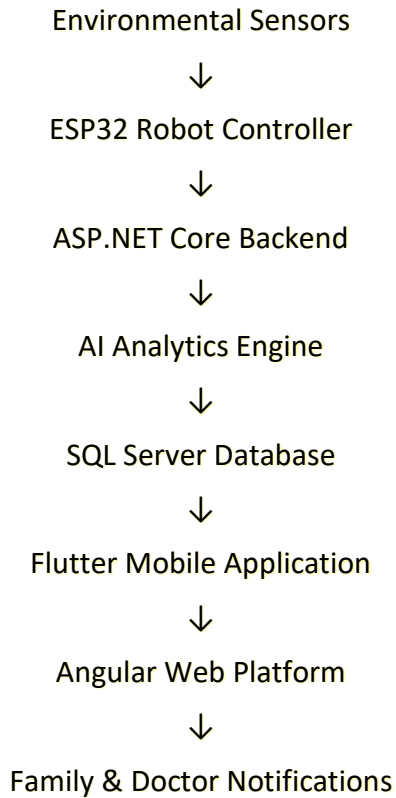
- Detecting unusual inactivity
- Identifying abnormal daily behavior
- Monitoring medication adherence
- Detecting possible fall situations
- Evaluating sleep quality
- Calculating wellness scores
- Predicting potential health risks
- Generating personalized recommendations
- Prioritizing emergency situations
- Supporting doctors in medical decision-making

Instead of relying on a single sensor, the AI combines data from multiple environmental sensors to improve accuracy and reduce false alarms.

System Workflow

All system components are connected through a secure cloud-based architecture.

The overall workflow is as follows:



This integrated architecture enables real-time monitoring, intelligent decision support, and continuous communication between elderly users, family members, and healthcare providers.